

Supplementary Information for

scAClc: A Multi-Objective Adaptive Clustering Framework for Single-Cell

Transcriptomics via Contrastive and Resolution-Aware Representation

Learning

Yucai Zheng¹, Rongqi Jin¹, Yajuan Zhang¹, Yue Cheng^{2*}, and Yunhe Wang^{1*}

¹ Hebei University of Technology, Tianjin, 300401, China

² Jilin University, Changchun, 130012, China

* Corresponding Author:

Yue Cheng, chengyue24@mails.jlu.edu.cn; Yunhe Wang, wangyh082@hebut.edu.cn

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Table S1. The statistics of 15 real scRNA-seq datasets.

No.	dataset	organ	platform	cell types	cells	gene number
1	10X_PBMC[1]	Human PBMC	10x Genomic	8	4271	16653
2	Human_p[2]	Human pancreas	inDrop	14	3605	20125
3	Mammary_Gland[3]	Mammary Gland	10x Genomic	7	4481	23341
4	Klein[4]	Mouse embryo stem cells	inDrop	4	2717	24047
5	Adam[5]	Kidney	Drop-seq	8	3660	23797
6	Chen [6]	Mouse brain	Drop-seq	46	12089	23284
7	Turtle_b[7]	Turtle brain	Drop-seq	15	18664	23500
8	Baron_human[8]	Human pancreas	inDrop	14	8569	20125
9	Muraro[9]	Pancreas	Smart-seq2	9	2122	19046
10	Plasschaert[10]	Trachea	inDrop	8	6977	28205
11	Pollen[11]	Human tissues	SMARTer	11	301	21721
12	Celseq1[12]	Human pancreas	CEL-seq	13	1004	34363
13	Celseq2[12]	Human pancreas	CEL-seq	13	2285	23341
14	QS_Diaphragm[13]	Diaphragm	Smart-seq2	5	870	23341
15	QS_Limb_Muscle[13]	Limb Muscle	Smart-seq2	6	1090	23341

Table S2. Clustering performance comparison of the different clustering algorithms on 15 real scRNA-seq datasets measured by ARI.

Dataset	scAClc	scMGCA	scTAG	scDFN	scDeepCluster	scSCC	scVI
10X_PBMC	0.7813	0.6704	0.6698	0.7672	0.7415	0.7057	0.6956
Human_p	0.9550	0.9024	0.6954	0.6298	0.5867	0.7406	0.7019
Mammary_Gland	0.9378	0.8910	0.8054	0.8861	0.7530	0.8223	0.8371
Klein	0.9108	0.6957	0.7739	0.7560	0.4559	0.7406	0.7253
Adam	0.9349	0.8635	0.8747	0.7411	0.7882	0.7553	0.5231
Chen	0.7880	0.5971	0.5508	0.6420	0.5396	0.3430	0.6362
Turtle_b	0.8926	0.4858	0.5590	0.4128	0.4653	0.7658	0.6873
Baron_human	0.9275	0.3798	0.3649	0.8034	0.4555	0.6786	0.6902
Muraro	0.9288	0.8858	0.8896	0.9280	0.6103	0.9003	0.7615
Plasschaert	0.9125	0.8215	0.8280	0.5337	0.2805	0.7465	0.5896
Pollen	0.8952	0.9267	0.7941	0.9335	0.8642	0.7317	0.1247
Celseq1	0.8274	0.5515	0.5274	0.5686	0.6644	0.4278	0.6302
Celseq2	0.9512	0.8244	0.7956	0.6974	0.6917	0.8617	0.7117
QS_Diaphragm	0.9538	0.9652	0.9153	0.9580	0.6561	0.4645	0.8072
QS_Limb_Muscle	0.9105	0.9751	0.9667	0.9779	0.6925	0.9559	0.7769
Average	0.9005	0.7624	0.7340	0.7490	0.6164	0.7094	0.6599

Table S3. Clustering performance comparison of the different clustering algorithms on 15 real scRNA-seq datasets measured by NMI.

Dataset	scAClc	scMGCA	scTAG	scDFN	scDeepCluster	scSCC	scVI
10X_PBMC	0.7989	0.7365	0.7730	0.7665	0.7704	0.7466	0.6063
Human_p	0.9109	0.8514	0.7525	0.8160	0.7723	0.7177	0.4564
Mammary_Gland	0.9164	0.8512	0.7786	0.8858	0.8210	0.8668	0.8173
Klein	0.9154	0.6978	0.7604	0.7386	0.5604	0.7177	0.7009
Adam	0.9153	0.8595	0.8710	0.8038	0.8238	0.8363	0.4323
Chen	0.7810	0.7651	0.7643	0.7655	0.7474	0.7219	0.2373
Turtle_b	0.8628	0.7064	0.6972	0.7175	0.7411	0.7843	0.4339
Baron_human	0.8818	0.6156	0.5900	0.8541	0.7410	0.8339	0.4511
Muraro	0.8847	0.8379	0.8406	0.8924	0.7569	0.8563	0.6475
Plasschaert	0.8618	0.7602	0.7749	0.6836	0.5452	0.7528	0.3894
Pollen	0.9339	0.9197	0.8925	0.9273	0.9300	0.8067	0.0082
Celseq1	0.8206	0.7590	0.7325	0.7569	0.7778	0.5610	0.3773
Celseq2	0.9175	0.8275	0.7980	0.8350	0.8427	0.8580	0.4714
QS_Diaphragm	0.9126	0.9392	0.8987	0.9448	0.8027	0.6494	0.6768
QS_Limb_Muscle	0.8599	0.9568	0.9448	0.9614	0.8153	0.9190	0.6596
Average	0.8782	0.8056	0.7913	0.8233	0.7632	0.7752	0.4910

Table S4. The number of cells on the Chen dataset and the correspondence between real cell type and cell number.

Cell type	Cluster number	Cell number
Glu1	0	13
Glu2	1	21
Glu3	2	13
Glu4	3	131
Glu5	4	200
Glu6	5	51
Glu7	6	211
Glu8	7	50
Glu9	8	35
Glu10	9	15
Glu11	10	42
Glu12	11	24
Glu13	12	24
Glu14	13	51
Glu15	14	25
GABA1	15	23
GABA2	16	27
GABA3	17	97
GABA4	18	18
GABA5	19	73
GABA6	20	49
GABA7	21	17
GABA8	22	402
GABA9	23	71
GABA10	24	23
GABA11	25	35
GABA12	26	62
GABA13	27	165
GABA14	28	71
GABA15	29	112
GABA16	30	66
GABA17	31	50
GABA18	32	31
Hista	33	17
POPC	34	51
OPC	35	1741
IMO	36	151
MO	37	3541
Astro	38	1148
Ependy	39	413
Tany	40	609
Epith1	41	818
Epith2	42	379
Micro	43	724
Macro	44	167

Table S5. Mapping Statistics Between Merged Clusters and Original Subtypes in Chen Dataset.

Cluster ID	Main Func Category	Main Category Ratio	Total Cells
0	Other (1-13)	0.995	1926
1	Other (1-13)	0.992	3545
2	GABA (1-18)	0.532	2411
3	Glu (1-15)	0.998	1034
4	Glu (1-15)	0.975	1211
5	Other (1-13)	0.998	829
6	Other (1-13)	0.889	682
7	Glu (1-15)	0.923	441

Table S6. Clustering results comparison of scAClc in different module ((i) only HVGs; (ii) only RFGs; (iii) without the hierarchical gene relevance module; (iv) with the hierarchical gene relevance module (scAClc)) measured by ARI.

Dataset	only HVGs	only RFGs	without	scAClc
10X_PBMC	0.7344	0.7141	0.5766	0.7813
Human_p	0.9554	0.8934	0.8973	0.9550
Mammary_Gland	0.9539	0.6858	0.8556	0.9378
Klein	0.8075	0.9123	0.8278	0.9108
Adam	0.8141	0.7929	0.8034	0.9349
Chen	0.6557	0.7733	0.8511	0.7880
Turtle_b	0.6935	0.7796	0.7663	0.8926
Baron_human	0.6723	0.9197	0.7949	0.9275
Muraro	0.9328	0.9331	0.7290	0.9288
Plasschaert	0.7544	0.7651	0.9361	0.9125
Pollen	0.8567	0.8656	0.3177	0.8952
Celseq1	0.9214	0.8889	0.6171	0.8274
Celseq2	0.9520	0.9256	0.7307	0.9512
QS_Diaphragm	0.9822	0.9519	0.5874	0.9538
QS_Limb_Muscle	0.7026	0.8068	0.6400	0.9105
Average	0.8259	0.8405	0.7287	0.9005

Table S7. Clustering results comparison of scAClc in different module ((i) only HVGs; (ii) only RFGs; (iii) without the hierarchical gene relevance module; (iv) with the hierarchical gene relevance module (scAClc)) measured by NMI.

Dataset	only HVGs	only RFGs	without	scAClc
10X_PBMC	0.7699	0.7656	0.7018	0.7989
Human_p	0.9127	0.8702	0.8777	0.9109
Mammary_Gland	0.9278	0.8281	0.8524	0.9164
Klein	0.8271	0.9160	0.8614	0.9154
Adam	0.8698	0.8484	0.8600	0.9153
Chen	0.6904	0.7652	0.7962	0.7810
Turtle_b	0.7428	0.8305	0.7590	0.8628
Baron_human	0.8284	0.8731	0.8341	0.8818
Muraro	0.8902	0.8904	0.8282	0.8847
Plasschaert	0.7550	0.7714	0.8713	0.8618
Pollen	0.9232	0.9078	0.5801	0.9339
Celseq1	0.8931	0.8774	0.7509	0.8206
Celseq2	0.9216	0.8859	0.8602	0.9175
QS_Diaphragm	0.9686	0.9067	0.8032	0.9126
QS_Limb_Muscle	0.7666	0.7538	0.7853	0.8599
Average	0.8458	0.8460	0.8015	0.8782

Table S8. Clustering results comparison of scAClc in different module ((i) standard contrastive learning; (ii) without contrastive learning; (iii) anchor-centered contrastive learning) measured by ARI.

Dataset	standard	without	anchor-centered
10X_PBMC	0.7649	0.7787	0.7813
Human_p	0.9545	0.9577	0.9550
Mammary_Gland	0.9082	0.9373	0.9378
Klein	0.9123	0.9170	0.9108
Adam	0.8159	0.9389	0.9349
Chen	0.8269	0.5490	0.7880
Turtle_b	0.7640	0.8651	0.8926
Baron_human	0.8023	0.8025	0.9275
Muraro	0.9309	0.9288	0.9288
Plasschaert	0.7872	0.9110	0.9125
Pollen	0.7756	0.8951	0.8952
Celseq1	0.6032	0.8250	0.8274
Celseq2	0.9534	0.9520	0.9512
QS_Diaphragm	0.9783	0.9574	0.9538
QS_Limb_Muscle	0.9211	0.9171	0.9105
Average	0.8466	0.8755	0.9005

Table S9. Clustering results comparison of scAClc in different module ((i) standard contrastive learning; (ii) without contrastive learning; (iii) anchor-centered contrastive learning) measured by NMI.

Dataset	standard	without	anchor-centered
10X_PBMC	0.7877	0.7966	0.7989
Human_p	0.9095	0.9138	0.9109
Mammary_Gland	0.8727	0.9163	0.9164
Klein	0.9173	0.9260	0.9154
Adam	0.8712	0.9189	0.9153
Chen	0.8028	0.6551	0.7810
Turtle_b	0.8270	0.8345	0.8628
Baron_human	0.8367	0.8364	0.8818
Muraro	0.8864	0.8846	0.8847
Plasschaert	0.7786	0.8594	0.8618
Pollen	0.8960	0.9339	0.9339
Celseq1	0.7618	0.9148	0.8206
Celseq2	0.9195	0.9175	0.9175
QS_Diaphragm	0.9528	0.9188	0.9126
QS_Limb_Muscle	0.8641	0.8654	0.8599
Average	0.8589	0.8728	0.8782

Table S10. Clustering results comparison of scAClc under different gene numbers of 2000, 3000, and 4000 screened by the HVG and RFG methods respectively measured by ARI.

Parameter	The gene number		
Dataset	2000	3000	4000
10X_PBMC	0.7544	0.7813	0.8089
Human_p	0.9559	0.9550	0.8939
Mammary_Gland	0.8321	0.9378	0.8198
Klein	0.9111	0.9108	0.8222
Adam	0.8129	0.9349	0.8588
Chen	0.7854	0.7880	0.7875
Turtle_b	0.7682	0.8926	0.7717
Baron_human	0.9228	0.9275	0.8028
Muraro	0.9316	0.9288	0.7345
Plasschaert	0.9126	0.9125	0.5971
Pollen	0.7686	0.8952	0.5967
Celseq1	0.8195	0.8274	0.8573
Celseq2	0.7588	0.9512	0.9540
QS_Diaphragm	0.5962	0.9538	0.9682
QS_Limb_Muscle	0.9273	0.9105	0.8408
Average	0.8305	0.9005	0.8076

Table S11. Clustering results comparison of scAClc under different gene numbers of 2000, 3000, and 4000 screened by the HVG and RFG methods respectively measured by NMI.

Parameter	The gene number		
Dataset	2000	3000	4000
10X_PBMC	0.7877	0.7989	0.8036
Human_p	0.9121	0.9109	0.8719
Mammary_Gland	0.8714	0.9164	0.8661
Klein	0.9134	0.9154	0.8505
Adam	0.8634	0.9153	0.8763
Chen	0.7736	0.7810	0.7825
Turtle_b	0.7495	0.8628	0.7388
Baron_human	0.8782	0.8818	0.8386
Muraro	0.8882	0.8847	0.8326
Plasschaert	0.8596	0.8618	0.6915
Pollen	0.8834	0.9339	0.8039
Celseq1	0.8132	0.8206	0.8410
Celseq2	0.8669	0.9175	0.9226
QS_Diaphragm	0.7923	0.9126	0.9339
QS_Limb_Muscle	0.8740	0.8599	0.8313
Average	0.8485	0.8782	0.8323

Table S12. Clustering results comparison of scAClc under different contrastive loss weight ranges of 0.10–0.06, 0.12–0.08, and 0.14–0.10 measured by ARI.

Parameter	The contrastive loss weight		
Dataset	0.10-0.06	0.12-0.08	0.14-0.10
10X_PBMC	0.7745	0.7813	0.7791
Human_p	0.9564	0.9550	0.9559
Mammary_Gland	0.9379	0.9378	0.6676
Klein	0.9150	0.9108	0.9106
Adam	0.9413	0.9349	0.9349
Chen	0.8309	0.7880	0.7753
Turtle_b	0.8310	0.8926	0.7951
Baron_human	0.9263	0.9275	0.9270
Muraro	0.9272	0.9288	0.7288
Plasschaert	0.9115	0.9125	0.6161
Pollen	0.8951	0.8952	0.8951
Celseq1	0.8274	0.8274	0.8274
Celseq2	0.9527	0.9512	0.9527
QS_Diaphragm	0.9538	0.9538	0.5857
QS_Limb_Muscle	0.9025	0.9105	0.9096
Average	0.8989	0.9005	0.8174

Table S13. Clustering results comparison of scAClc under different contrastive loss weight ranges of 0.10–0.06, 0.12–0.08, and 0.14–0.10 measured by NMI.

Parameter	The contrastive loss weight		
Dataset	0.10-0.06	0.12-0.08	0.14-0.10
10X_PBMC	0.7954	0.7989	0.7949
Human_p	0.9122	0.9109	0.9110
Mammary_Gland	0.9164	0.9164	0.8242
Klein	0.9222	0.9154	0.9150
Adam	0.9217	0.9153	0.9165
Chen	0.8055	0.7810	0.7634
Turtle_b	0.8017	0.8628	0.8249
Baron_human	0.8810	0.8818	0.8814
Muraro	0.8835	0.8847	0.8277
Plasschaert	0.8595	0.8618	0.6991
Pollen	0.9339	0.9339	0.9339
Celseq1	0.8206	0.8206	0.8206
Celseq2	0.9186	0.9175	0.9186
QS_Diaphragm	0.9125	0.9126	0.7686
QS_Limb_Muscle	0.8518	0.8599	0.8618
Average	0.8758	0.8782	0.8441

Table S14. Clustering results comparison of scAClc under different initial clustering resolutions of 1.0, 2.0, and 4.0 measured by ARI.

Parameter	The initial clustering resolution		
Dataset	1.0	2.0	4.0
10X_PBMC	0.7484	0.7813	0.7792
Human_p	0.9580	0.9550	0.9570
Mammary_Gland	0.9328	0.9378	0.6771
Klein	0.9119	0.9108	0.9143
Adam	0.8717	0.9349	0.9358
Chen	0.7339	0.7880	0.8310
Turtle_b	0.6435	0.8926	0.7531
Baron_human	0.8107	0.9275	0.7931
Muraro	0.7336	0.9288	0.7288
Plasschaert	0.9117	0.9125	0.9083
Pollen	0.8952	0.8952	0.7177
Celseq1	0.8255	0.8274	0.6230
Celseq2	0.9530	0.9512	0.9534
QS_Diaphragm	0.9595	0.9538	0.9556
QS_Limb_Muscle	0.5565	0.9105	0.9109
Average	0.8297	0.9005	0.8292

Table S15. Clustering results comparison of scAClc under different initial clustering resolutions of 1.0, 2.0, and 4.0 measured by NMI.

Parameter	The initial clustering resolution		
Dataset	1.0	2.0	4.0
10X_PBMC	0.7797	0.7989	0.7984
Human_p	0.9138	0.9109	0.9125
Mammary_Gland	0.9089	0.9164	0.8280
Klein	0.9162	0.9154	0.9223
Adam	0.8888	0.9153	0.9154
Chen	0.7392	0.7810	0.8065
Turtle_b	0.7365	0.8628	0.7625
Baron_human	0.8512	0.8818	0.8295
Muraro	0.8301	0.8847	0.8283
Plasschaert	0.8596	0.8618	0.8572
Pollen	0.9339	0.9339	0.8481
Celseq1	0.8144	0.8206	0.7720
Celseq2	0.9201	0.9175	0.9195
QS_Diaphragm	0.9232	0.9126	0.9182
QS_Limb_Muscle	0.7518	0.8599	0.8613
Average	0.8512	0.8782	0.8520

Table S16. Clustering results comparison of scAClc under different tolerance thresholds of 0.01, 0.05, and 0.10 measured by ARI.

Parameter	The tolerance threshold		
Dataset	0.01	0.05	0.10
10X_PBMC	0.7861	0.7813	0.7862
Human_p	0.9180	0.9550	0.8966
Mammary_Gland	0.6956	0.9378	0.9379
Klein	0.9136	0.9108	0.9139
Adam	0.9481	0.9349	0.9349
Chen	0.6390	0.7880	0.8410
Turtle_b	0.6292	0.8926	0.6965
Baron_human	0.5089	0.9275	0.9275
Muraro	0.7376	0.9288	0.7308
Plasschaert	0.4452	0.9125	0.9116
Pollen	0.8202	0.8952	0.8952
Celseq1	0.8892	0.8274	0.8251
Celseq2	0.7757	0.9512	0.9512
QS_Diaphragm	0.9673	0.9538	0.9595
QS_Limb_Muscle	0.8786	0.9105	0.9255
Average	0.7702	0.9005	0.8756

Table S17. Clustering results comparison of scAClc under different tolerance thresholds of 0.01, 0.05, and 0.10 measured by NMI.

Parameter	The tolerance threshold		
Dataset	0.01	0.05	0.10
10X_PBMC	0.7960	0.7989	0.8036
Human_p	0.9050	0.9109	0.8748
Mammary_Gland	0.8088	0.9164	0.9164
Klein	0.9204	0.9154	0.9203
Adam	0.9286	0.9153	0.9153
Chen	0.7100	0.7810	0.8140
Turtle_b	0.7035	0.8628	0.7552
Baron_human	0.7716	0.8818	0.8818
Muraro	0.8418	0.8847	0.8292
Plasschaert	0.6276	0.8618	0.8580
Pollen	0.8967	0.9339	0.9339
Celseq1	0.8746	0.8206	0.8142
Celseq2	0.8841	0.9175	0.9175
QS_Diaphragm	0.9347	0.9126	0.9232
QS_Limb_Muscle	0.8135	0.8599	0.8774
Average	0.8278	0.8782	0.8690

Table S18. Clustering results comparison of scAClc under different merging trigger thresholds of $\bar{d}/4$, $\bar{d}/2$, and $\bar{d}$ measured by ARI.

Parameter	The merging trigger threshold		
Dataset	$\bar{d}/4$	$\bar{d}/2$	$\bar{d}$
10X_PBMC	0.6470	0.7813	0.3272
Human_p	0.4709	0.9550	0.2590
Mammary_Gland	0.3707	0.9378	0.4113
Klein	0.6606	0.9108	0.5007
Adam	0.3764	0.9349	0.2468
Chen	0.8898	0.7880	0.1841
Turtle_b	0.4865	0.8926	0.2972
Baron_human	0.2954	0.9275	0.2720
Muraro	0.5853	0.9288	0.3759
Plasschaert	0.2821	0.9125	0.7369
Pollen	0.8384	0.8952	0.0755
Celseq1	0.2705	0.8274	0.3973
Celseq2	0.3724	0.9512	0.3535
QS_Diaphragm	0.5961	0.9538	0.5154
QS_Limb_Muscle	0.4773	0.9105	0.4172
Average	0.5080	0.9005	0.3580

Table S19. Clustering results comparison of scAClc under different merging trigger thresholds of $\bar{d}/4$, $\bar{d}/2$, and $\bar{d}$ measured by NMI.

Parameter	The merging trigger threshold		
Dataset	$\bar{d}/4$	$\bar{d}/2$	$\bar{d}$
10X_PBMC	0.7303	0.7989	0.5119
Human_p	0.7562	0.9109	0.4506
Mammary_Gland	0.7162	0.9164	0.5425
Klein	0.8048	0.9154	0.6638
Adam	0.7044	0.9153	0.4508
Chen	0.8226	0.7810	0.3469
Turtle_b	0.7775	0.8628	0.4501
Baron_human	0.6819	0.8818	0.4567
Muraro	0.7436	0.8847	0.5087
Plasschaert	0.5618	0.8618	0.6994
Pollen	0.9227	0.9339	0.3017
Celseq1	0.6330	0.8206	0.5197
Celseq2	0.7222	0.9175	0.4886
QS_Diaphragm	0.7938	0.9126	0.6164
QS_Limb_Muscle	0.7113	0.8599	0.5384
Average	0.7388	0.8782	0.5031

Table S20. Clustering results comparison of scAClc under different merging trigger thresholds of $\bar{d}/4$, $\bar{d}/2$, and $\bar{d}$ measured by training epochs.

Parameter	The merging trigger threshold		
Dataset	$\bar{d}/4$	$\bar{d}/2$	$\bar{d}$
10X_PBMC	36	10	18
Human_p	56	9	17
Mammary_Gland	45	13	21
Klein	64	10	16
Adam	180	15	26
Chen	15	9	20
Turtle_b	44	17	24
Baron_human	27	14	22
Muraro	43	11	21
Plasschaert	94	15	17
Pollen	7	5	17
Celseq1	56	16	21
Celseq2	108	11	21
QS_Diaphragm	20	12	16
QS_Limb_Muscle	62	13	17
Average	57	12	20

Table S21. Clustering results comparison of scAClc under different merging trigger thresholds of $\bar{d}/4$, $\bar{d}/2$, and $\bar{d}$ measured by number of cluster merges.

Parameter	The merging trigger threshold		
Dataset	$\bar{d}/4$	$\bar{d}/2$	$\bar{d}$
10X_PBMC	6	3	8
Human_p	4	3	7
Mammary_Gland	4	6	10
Klein	5	3	6
Adam	10	7	13
Chen	6	4	10
Turtle_b	6	5	9
Baron_human	3	7	11
Muraro	4	5	10
Plasschaert	7	3	4
Pollen	2	2	8
Celseq1	4	8	11
Celseq2	7	5	10
QS_Diaphragm	4	3	5
QS_Limb_Muscle	6	3	5
Average	5	4	8

Table S22. Clustering results comparison of scAClc under different merging trigger thresholds of $\bar{d}/4$, $\bar{d}/2$, and $\bar{d}$ measured by total training time.

Parameter	The merging trigger threshold		
Dataset	$\bar{d}/4$	$\bar{d}/2$	$\bar{d}$
10X_PBMC	263	143	168
Human_p	213	123	138
Mammary_Gland	195	164	167
Klein	145	95	101
Adam	275	124	134
Chen	526	442	468
Turtle_b	887	718	732
Baron_human	438	340	342
Muraro	86	67	71
Plasschaert	381	279	287
Pollen	12	9	10
Celseq1	57	40	43
Celseq2	108	69	73
QS_Diaphragm	31	29	28
QS_Limb_Muscle	57	37	38
Average	245	178	186

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